

3. Correlation Analysis (Detail Graph refer to Graph in Appendix C)

3.1 Global Correlation Analysis

For global correlation analysis, Pearson, Kendall's Tau, Spearman, and Mutual Information were used as the primary methods. Python serves as the main tool for all calculations and visual generation.

Figure 3.1 Heatmap of Pearson Correlation

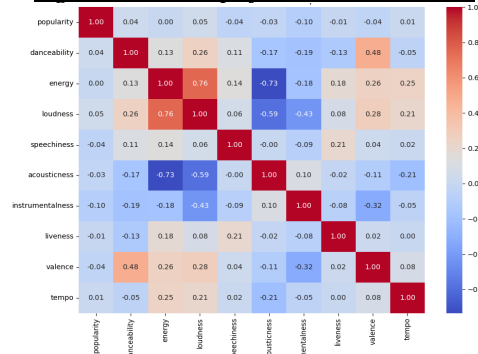


Figure 3.2 Heatmap of Kendall's Tau Correlation

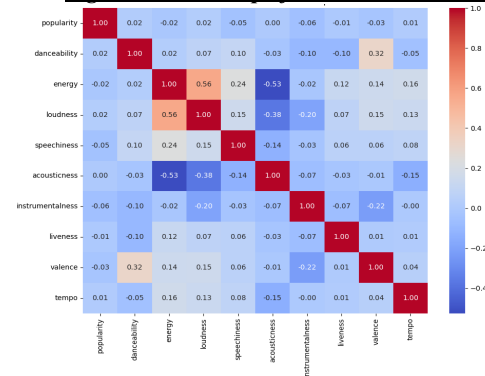
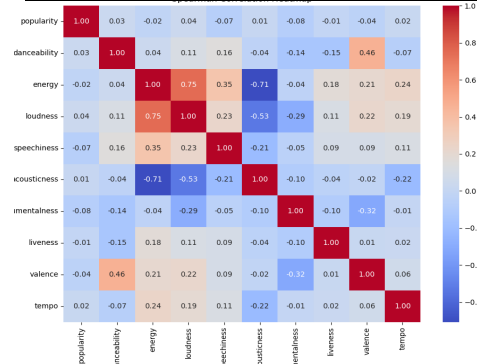


Figure 3.3 Heatmap of Spearman Correlation

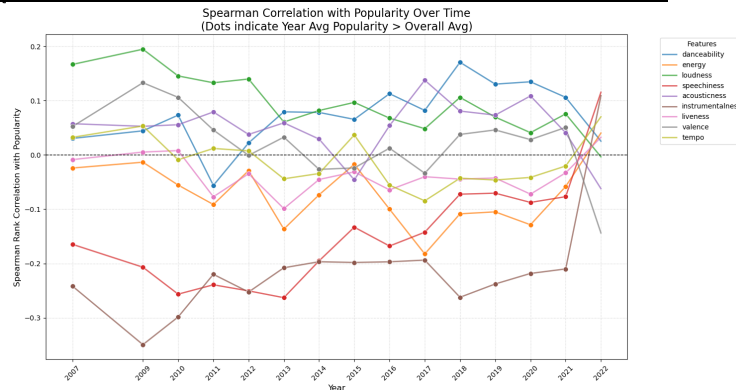


The three heatmaps above show the global correlations between all continuous and ratio attributes, calculated by using the Pearson, Kendall's Tau, and Spearman methods respectively. To highlight the key findings, energy and loudness have the highest correlation (around 0.56 to 0.76), while energy and acousticness have the lowest correlation across all methods (around -0.53 to -0.71). It appears that a song with high energy tends to be loud but low in acousticness.

3.2 Popularity Analysis

In this section, we analyze the factors driving song popularity on Spotify. We focus on numerical audio attributes and conduct our analysis through temporal trends and high-versus-low popularity comparisons.

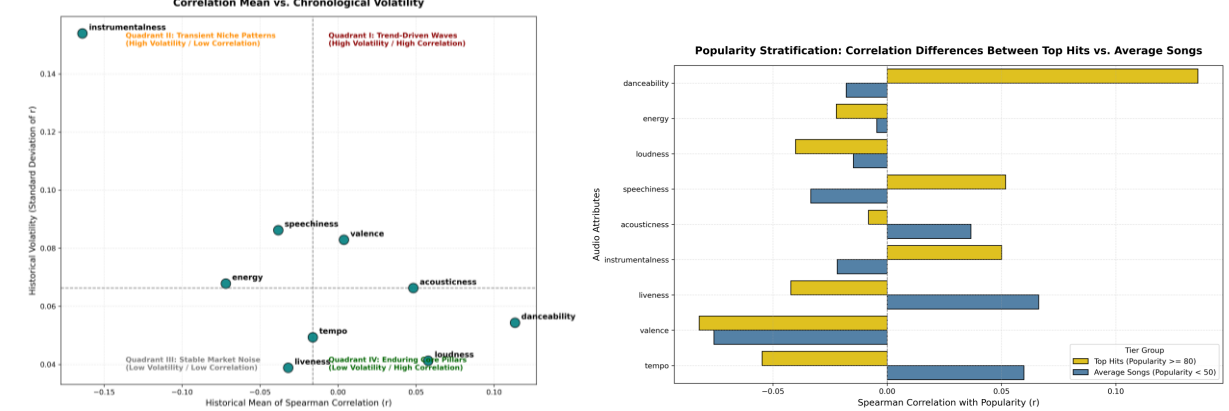
Figure 3.4 Line Chart of Spearman Correlation with Popularity Over Time (Data Selection and Detail reason for Methodology refer to Data Selection and Selection of Methodology in Appendix C)



The line chart above shows how the Spearman correlation between popularity and other features changes over time. Since 2010, the correlation for speechiness has gone up and became positive number in 2022. This change might be because rap music is becoming more popular. Besides, danceability went up between 2011 and 2018, which might be connected to the success of Hip-

Hop, K-Pop, and short video platforms. Surprisingly, in 2022, the correlation of instrumentality has an interesting jump, while loudness, valance, danceability and acoustiness drop quickly.

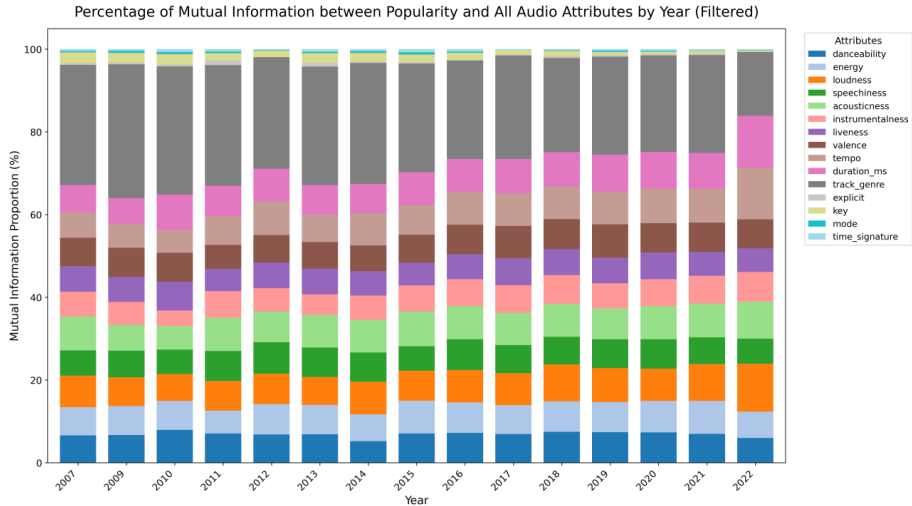
Figure 3.5 4-Quadrant Chart of Music Features (2007-2022) Figure 3.6 Side-by-Side Bar Chart of Popularity Stratification



To make it straightforward, we connect the volatility with the mean of correlation in a quadrant graph, where attributes can be divided into four groups. From this, we can easily see if an attribute is related and stable, or if it changes easily. For example, instrumentality stays in Quadrant II. This shows that even though it jumped in 2022, its average correlation over the years is still very low. Therefore, it seems to be a short-term trend, not a long-term evergreen feature. Conversely, although danceability decrease between 2018 to 2022, it is still relatively stable and might be a long-term evergreen feature. On the other hand, by focusing on the difference in correlation between top hits and average songs, the former seems to be focusing more on danceability, while the latter focusing more on liveness. Interestingly, six of the attributes (danceability, speechiness, acoustiness, instrumentality, liveness, tempo) show an inverse relationship between the two groups.

3.3 Mutual Information Insights

Figure 3.7 Stacked Bar Chart of Mutual Information Proportion by Year (Detail data refer to Detailed Mutual Information Data in Appendix C)

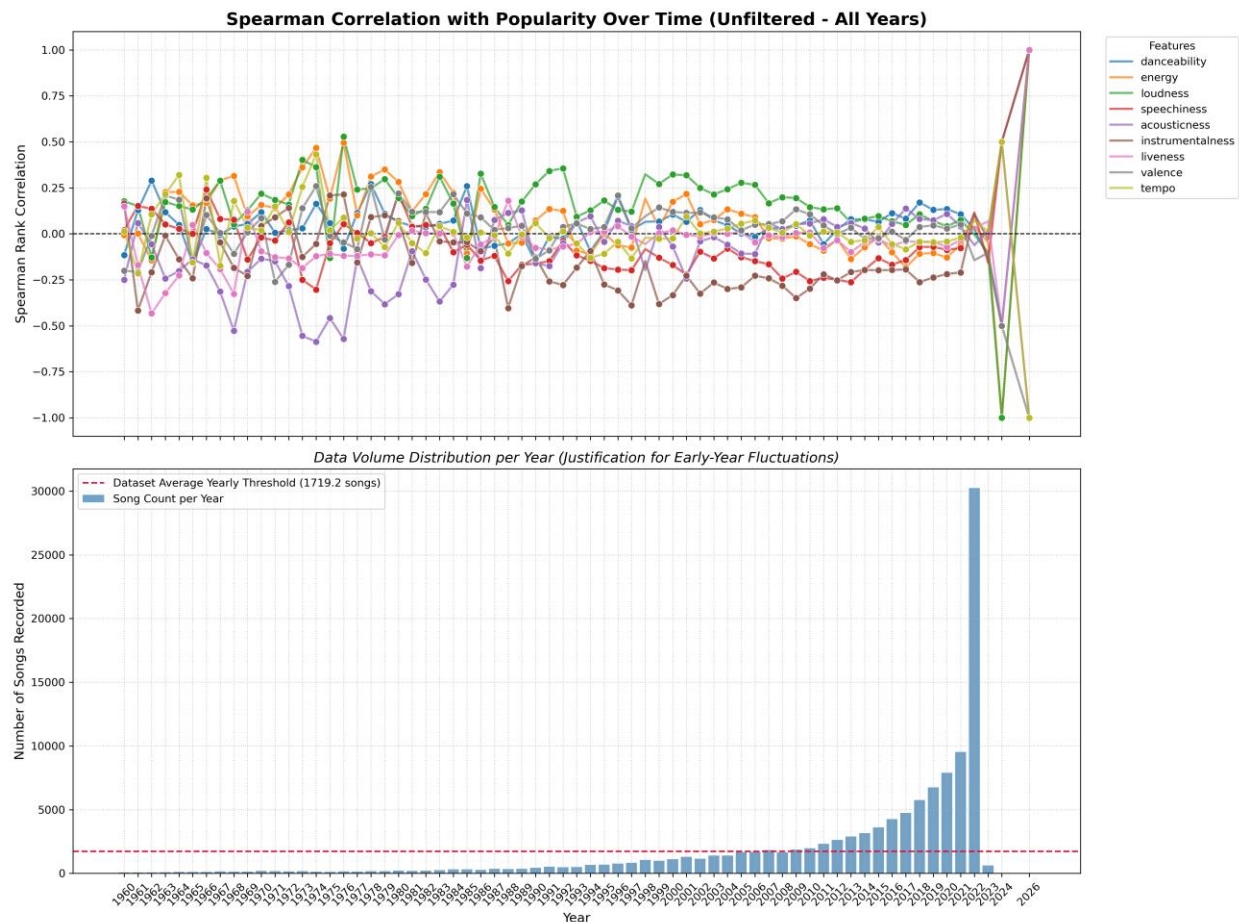


When we focus on mutual information, which analyzes non-numeric attributes out of the linear relationship assumption, genre has the highest mutual information each year. This might be the reason why the positive correlation with popularity calculated previously was quite low (around 0.50). Using mutual information shows that genre seems to be a very important factor linked to a song's popularity. It also shows that duration, tempo, and loudness are becoming stronger indicators of a song's popularity.

Appendix C – Correlation Analysis

Data Selection

The dataset we used collected songs published from 1960 to 2026. However, due to the collection preferences of the dataset, certain years have relatively few recorded songs. Therefore, to eliminate minor fluctuations and noise caused by low sample size, we applied a formula ($\text{Selected Year} \geq \text{Total Number of Songs} / \text{Total Number of Unique Years}$) to filter and finally select 2007, 2009-2022 for the main analysis. The detailed line chart and bar chart are provided below for reference.



Selection of Methodology

In the popularity analysis (Section 3.2), we used Spearman correlation to calculate the relationship between popularity and other music attributes. We did this because music features rarely follow a normal distribution. Since Spearman is a nonparametric method, it does not make assumptions about how the data is distributed, making our results more accurate. Additionally, by focusing on popularity rankings instead of raw numbers, Spearman correlation ignores the

massive scale of extreme values and protects the overall trend. For these reasons, we chose Spearman over Pearson and Kendall's Tau.

Detailed Mutual Information Data

The table below presents the precise numerical values of the Mutual Information proportions discussed in Section 3.3 (Mutual Information Insights).

	2007	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022
danceability	6.61	6.69	7.94	7.05	6.84	6.85	5.19	7.07	7.2	6.9	7.49	7.4	7.32	6.97	6
energy	6.8	7.01	6.99	5.53	7.33	7.13	6.51	7.92	7.38	7.04	7.38	7.29	7.61	7.96	6.36
loudness	7.66	6.94	6.49	7.21	7.36	6.78	7.91	7.24	7.81	7.72	8.88	8.18	7.81	8.95	11.6
speechiness	6.09	6.47	5.92	7.25	7.64	7.09	7.02	5.96	7.44	6.79	6.68	6.96	7.09	6.44	6.01
acousticness	8.21	6.2	5.64	8.03	7.36	7.9	7.9	8.37	8.05	7.77	8.01	7.49	7.98	8.16	8.97
instrumentalness	5.98	5.58	3.81	6.43	5.66	4.95	5.9	6.32	6.49	6.71	6.94	6.05	6.56	6.75	7.19
liveness	6.16	6.05	6.97	5.38	6.24	6.23	5.88	5.5	6.04	6.48	6.28	6.21	6.47	5.78	5.69
valence	6.89	7.05	7.03	5.78	6.63	6.46	6.22	6.78	7.11	7.85	7.23	8.06	7.08	7.08	7.01
tempo	6.22	5.79	5.47	6.87	8.12	6.64	7.82	7.21	7.99	7.82	7.93	7.84	8.4	8.23	12.52
duration_ms	6.52	6.24	8.58	7.45	7.91	7.15	7.1	7.92	7.97	8.38	8.29	9.06	8.83	8.61	12.57
track_genre	29.07	32.35	31	29.18	26.96	28.63	29.25	26.26	23.79	25.02	22.78	23.69	23.34	23.63	15.37
explicit	0.39	0.39	0.47	1.1	0	0.89	0.3	0.33	0.16	0.11	0.43	0.33	0.29	0.33	0.13
key	2.65	2.3	2.48	1.81	1.53	2.36	2.12	1.86	1.69	1.07	1.27	0.77	0.59	0.71	0.32
mode	0.27	0.51	0.5	0.34	0.16	0.28	0.39	0.49	0.28	0.02	0.15	0.25	0.16	0.06	0.08
time_signature	0.48	0.42	0.71	0.6	0.25	0.66	0.51	0.79	0.59	0.3	0.25	0.42	0.46	0.34	0.17

Graph

Figure 3.1 Heatmap of Pearson Correlation

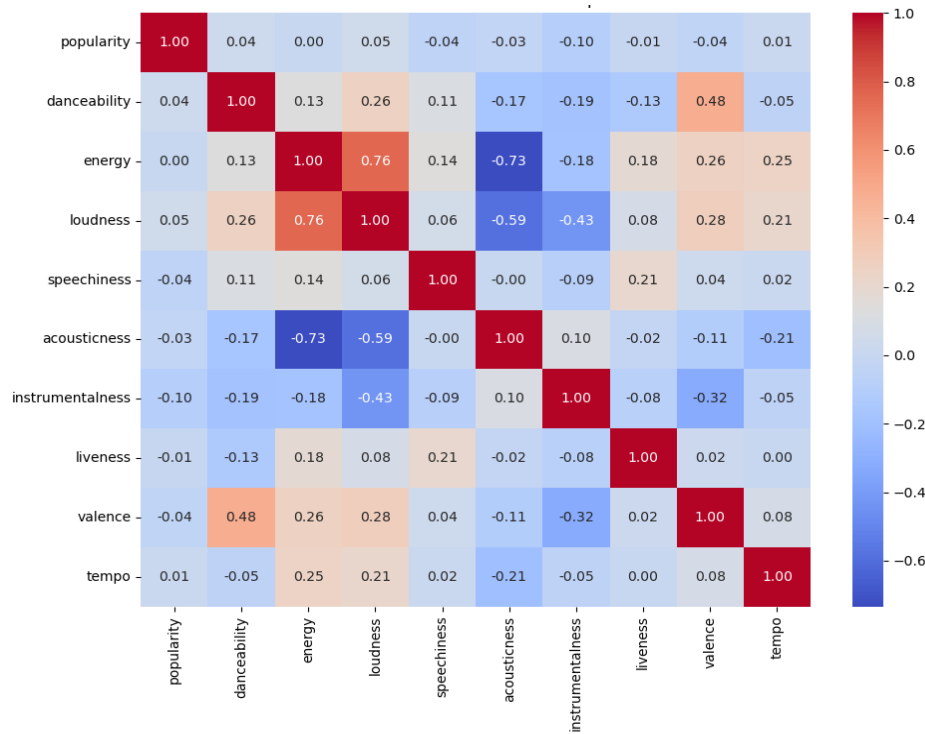


Figure 3.2 Heatmap of Kendall's Tau Correlation

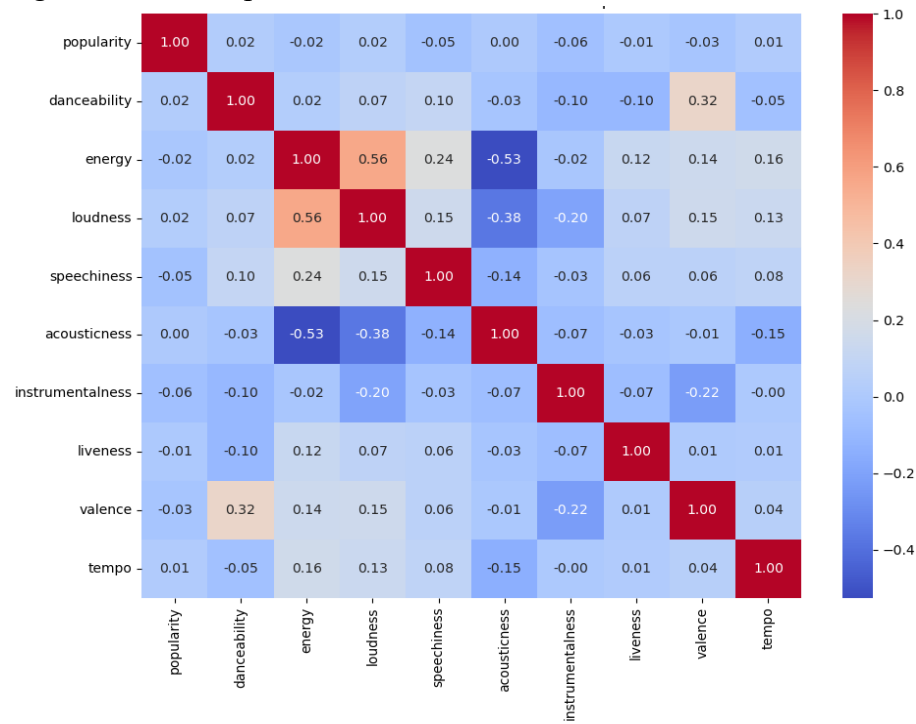


Figure 3.3 Heatmap of Spearman Correlation

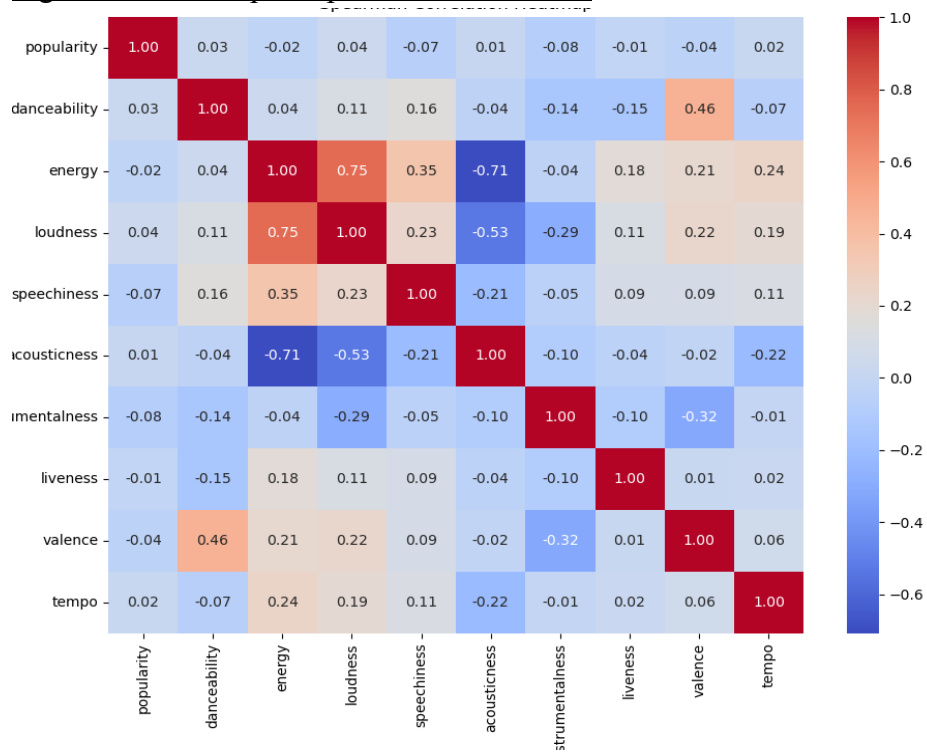


Figure 3.4 Line Chart of Spearman Correlation with Popularity

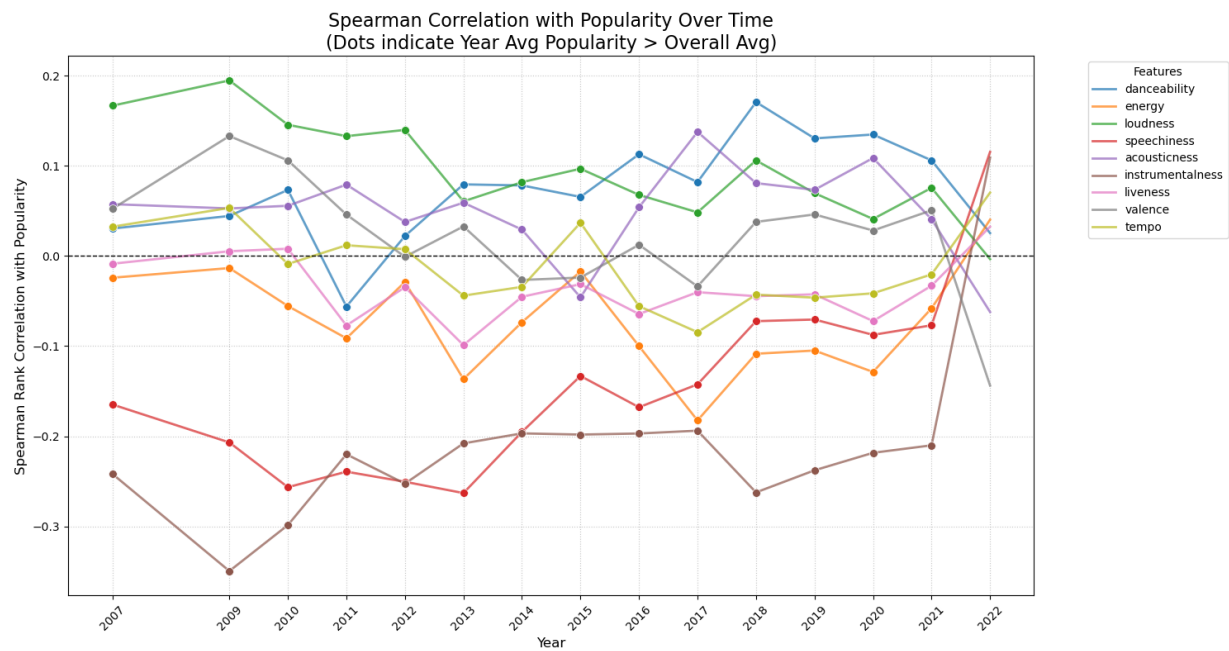


Figure 3.5 4-Quadrant Chart of Music Features (2007-2022)

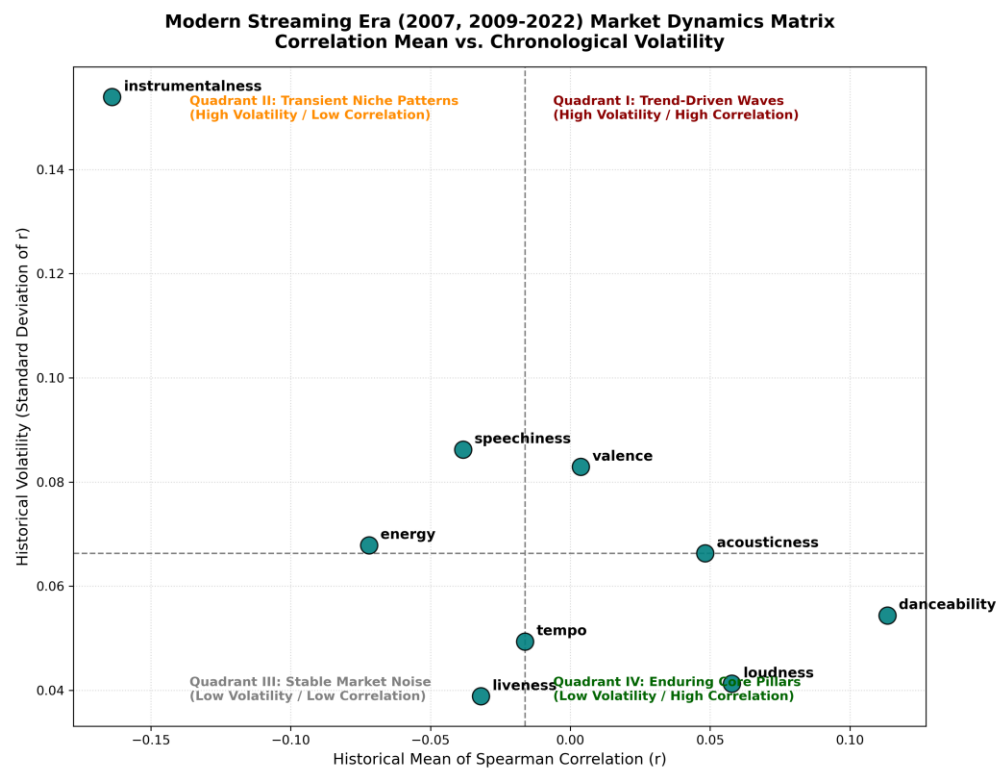


Figure 3.6 Side-by-Side Bar Chart of Popularity Stratification

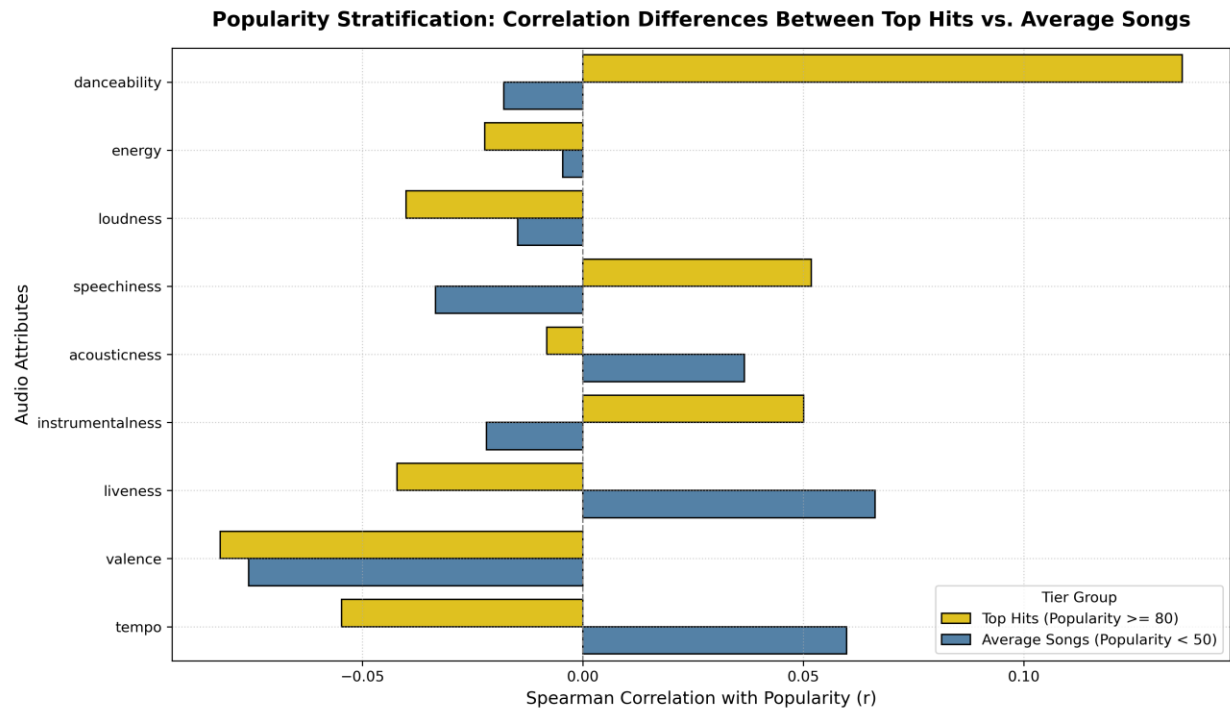


Figure 3.7 Stacked Bar Chart of Mutual Information Proportion by Year

